

Grassmann-perturbed hydrofoil polar sweep

10,000 PGA-perturbed foil shapes, swept over angle of attack, Reynolds number and transition N_{crit} with XFOil. Completed 25 Jul 2026, 01:45.

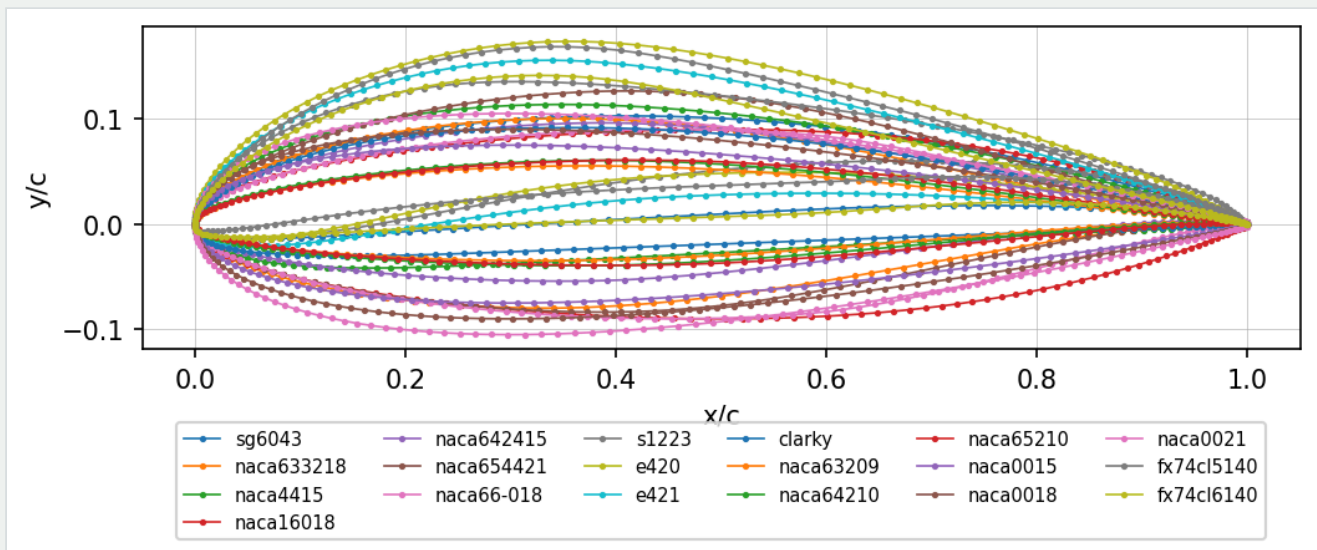
10,000	21,700,000	89.5%	92.4%
PERTURBED SHAPES	XFOIL RUNS	CONVERGED	MEDIAN PER-SHAPE CONVERGENCE

00 Baseline shapes

19 UIUC/AeroSandbox airfoils were selected to span the tidal/marine design space: low-Re performance, high Cl/Cd , thin cross-flow sections and cavitation-benchmark geometries. Each was repanelled to 100 points per side (enabled) so every shape shares a common landmark count for the Grassmann embedding below.

SHAPE	RATIONALE (FROM CONFIG)
sg6043	low Re performance: Re 1e5-1e6, best cavitation margin
naca633218	high Cl/Cd : Re 5e5-3e6, highest Cl/Cd
naca4415	robust deployment: Re 3e5-2e6, DOE RM1 reference
naca16018	low cavitation inception: Re 5e5-5e6, marine/propeller
naca642415	full-scale tidal: Re 5e5-6e6, passive-pitch reference
naca654421	65-series, passive-pitch studies
naca66-018	cavitation validation benchmark
s1223	high-lift, lower TSR machines
e420	Eppler, cited in tidal parametric studies

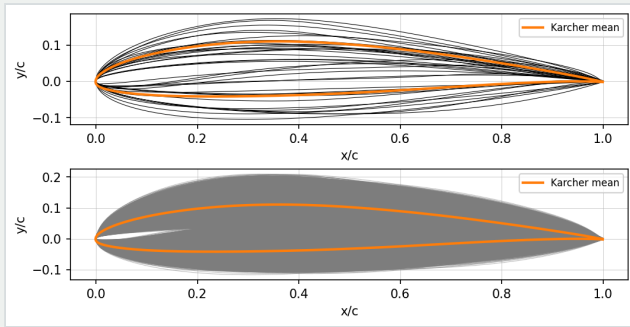
SHAPE	RATIONALE (FROM CONFIG)
e421	Eppler, cited in tidal parametric studies
clarky	cavitation tunnel validation baseline
naca63209	thinner variant for tip sections
naca64210	thinner variant for tip sections
naca65210	thinner variant for tip sections
naca0015	cross-flow / Darrieus tidal rotors
naca0018	cross-flow / Darrieus tidal rotors
naca0021	cross-flow / Darrieus tidal rotors
fx74cl5140	Wortmann, high-lift low-Re
fx74cl6140	Wortmann, high-lift low-Re



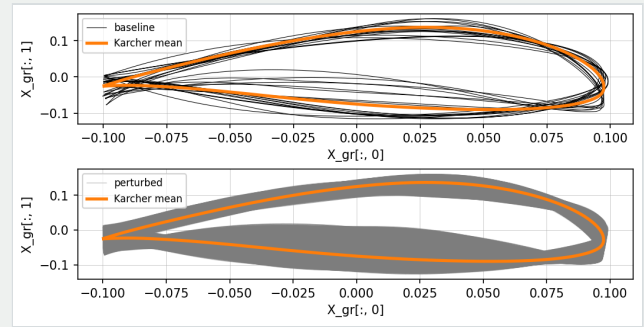
All 19 repanelled shapes, overlaid.

01 Grassmann / PGA shape perturbation

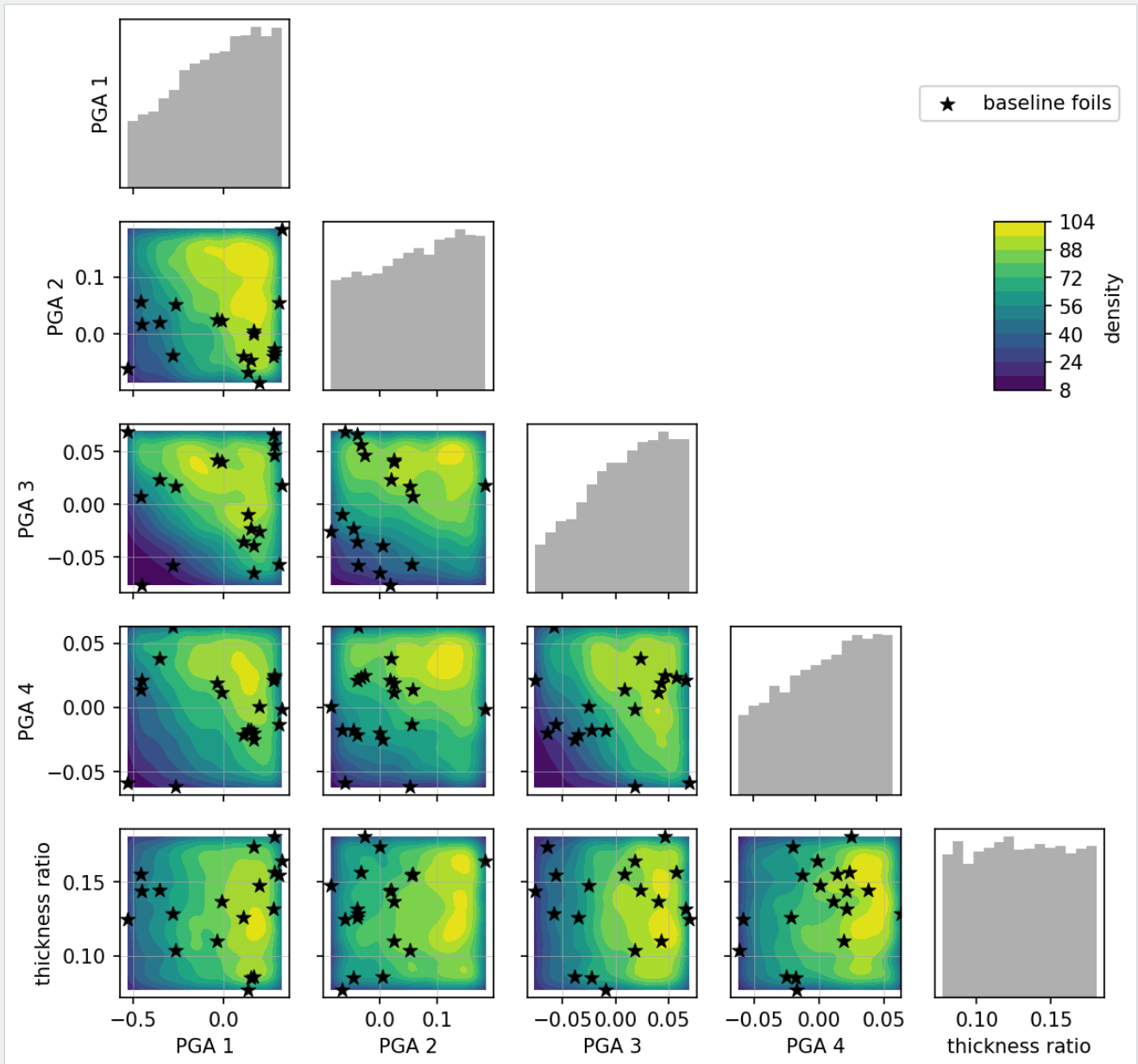
The baseline shapes are embedded on the Grassmannian, and a principal geodesic analysis (PGA) basis with $r = 4$ modes is fit across them (Doronina et al. 2022 truncation). Sampling that basis 10,000 times (seed 0) produces the perturbed-shape population the sweep actually runs over — not the 19 baseline foils directly.



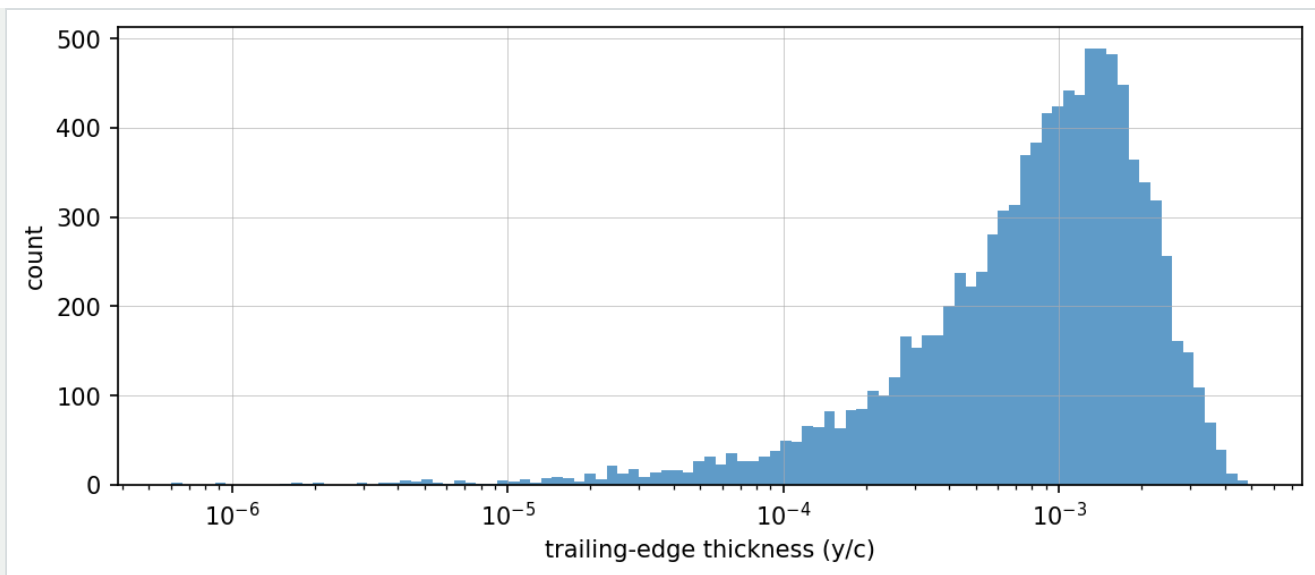
Perturbed outlines in physical (x/c , y/c) space.



Same samples in Grassmann tangent-space coordinates.



Pairwise distribution of the 4 PGA coordinates plus thickness ratio across all 10,000 samples, baseline shapes marked.



Trailing-edge thickness distribution across perturbed samples — a sanity check that sampling did not produce self-intersecting or degenerate trailing edges.

02 Sweep configuration

Operating conditions

Angle of attack	-10° to 20°, step 1° (31 pts)
Reynolds number	5 values, $5e+05$ – $1e+07$
N_{crit}	14 values, 1–13 (+1 forced)
Forced transition	Tu 5.0%, x/c trip 0.05 / 0.05

Cavitation reference

Chord	0.5 m
Depth	30 m
Temperature	10°C

XFoil

max_iter	300
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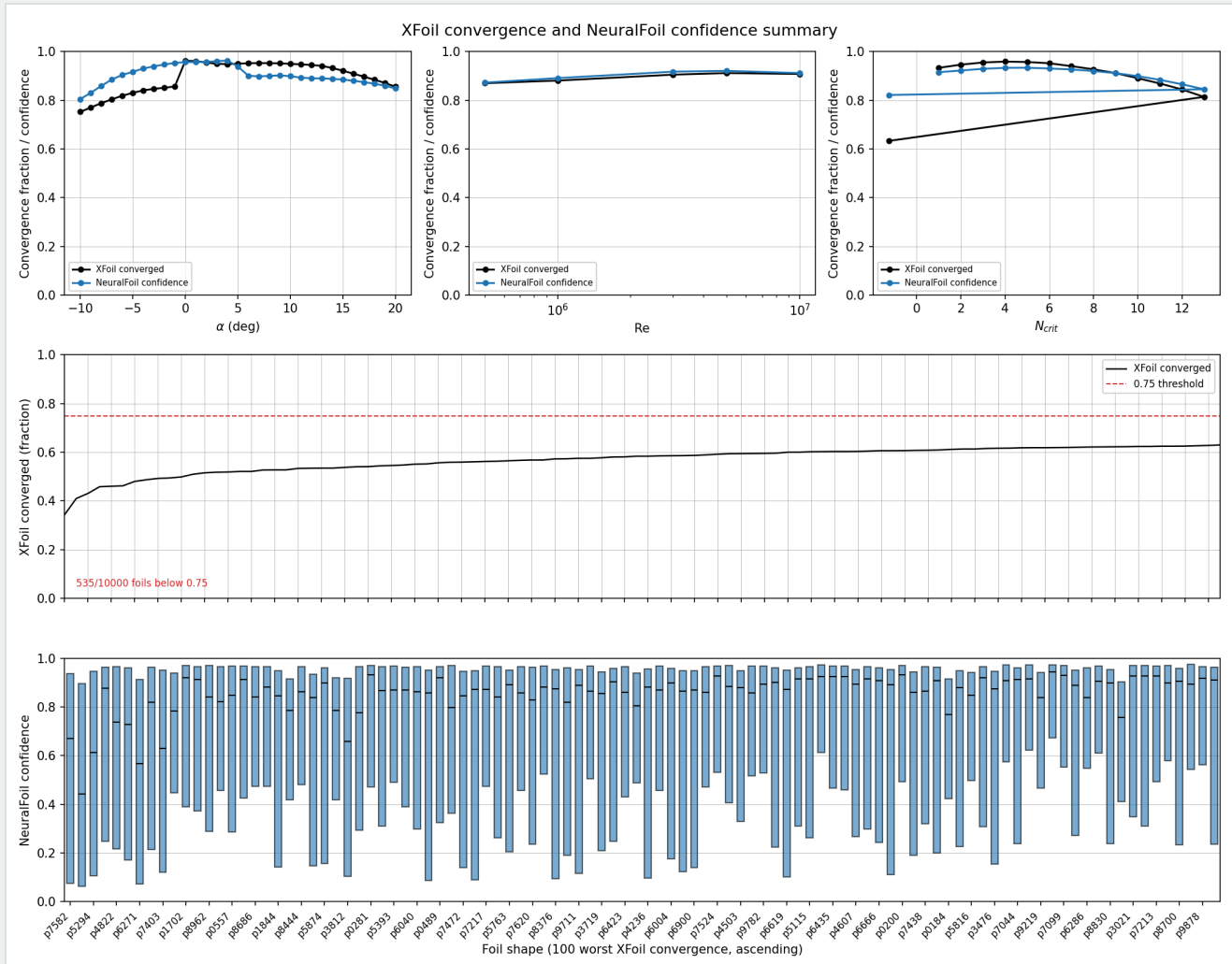
Total runs

shapes $\times \alpha \times Re \times N_{crit}$	21,700,000
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Output written to `foilpolars.nc` (2.1 GB) and a flattened `foilpolars.csv` (10.6 GB), checkpointed every 250 foils during the run.

03 XFoil convergence

Across all 21,700,000 runs, XFoil converged on 89.5% of points, with a per-shape median of 92.4%. 535 of 10,000 shapes (5.3%) converge on fewer than 75% of their (α , Re, N_{crit}) points — the geometrically extreme samples flagged below.



Top row: convergence marginalised over α , Re and N_{crit} in turn (NeuralFoil confidence shown alongside for reference). Middle: XFoil convergence per shape, worst-first. Bottom: NeuralFoil confidence distribution per shape.

04 Worst-converging shapes

Lowest XFoil convergence

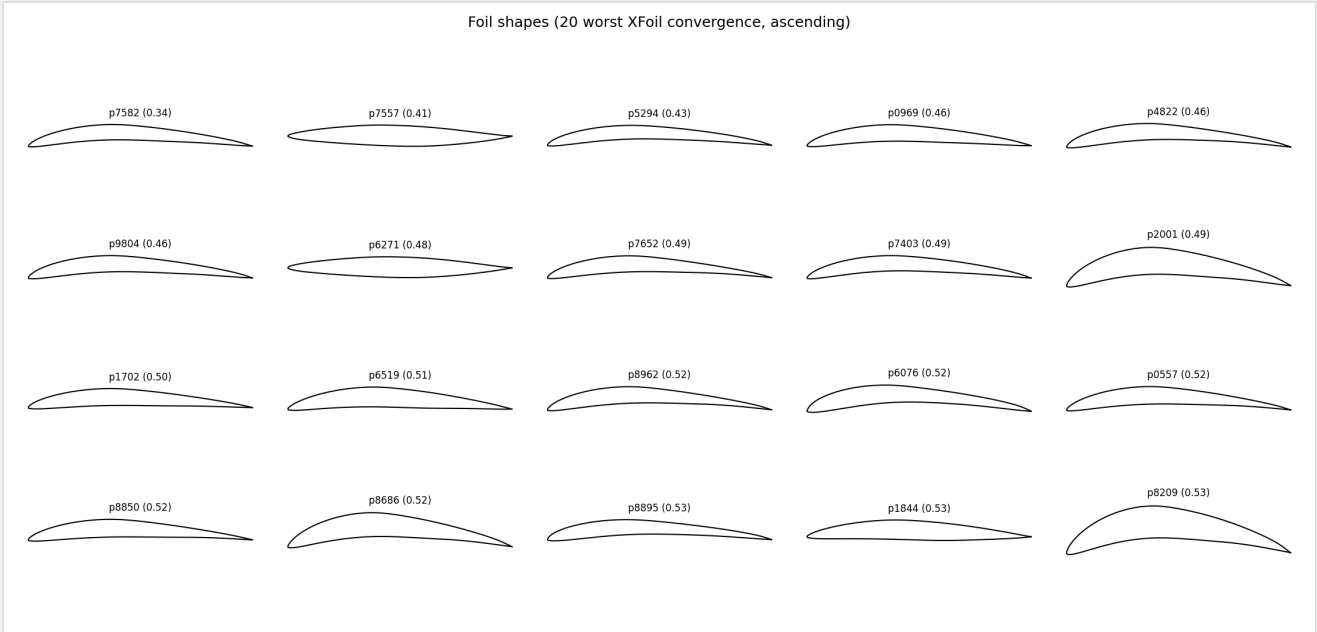
SHAPE	CONVERGED
p7582	34.4%
p7557	41.0%
p5294	43.1%

Highest XFoil convergence

SHAPE	CONVERGED
p5940	98.8%
p7491	98.3%
p8071	98.2%

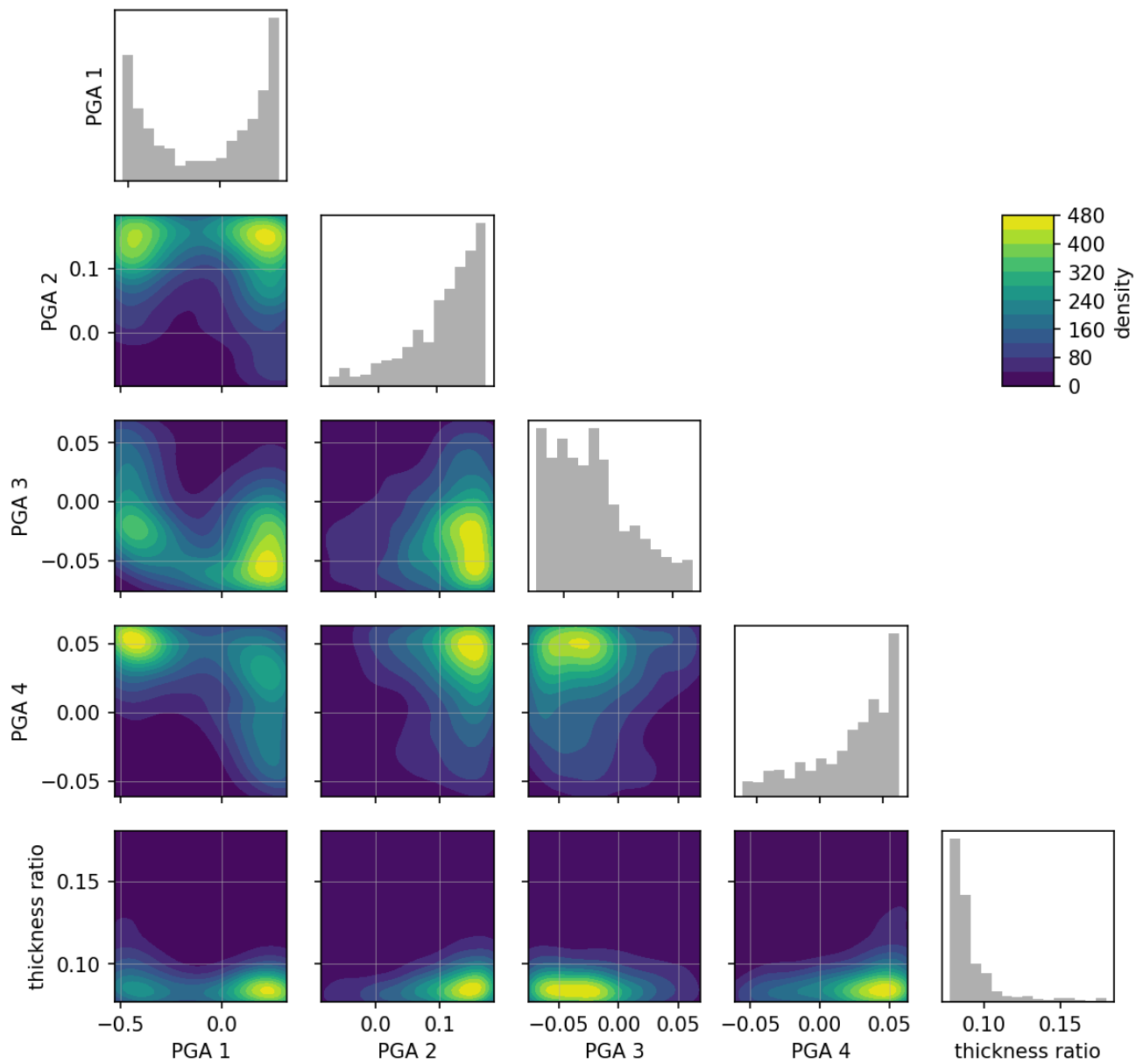
SHAPE	CONVERGED
p0969	45.9%
p4822	46.0%

SHAPE	CONVERGED
p2423	98.2%
p6048	98.2%



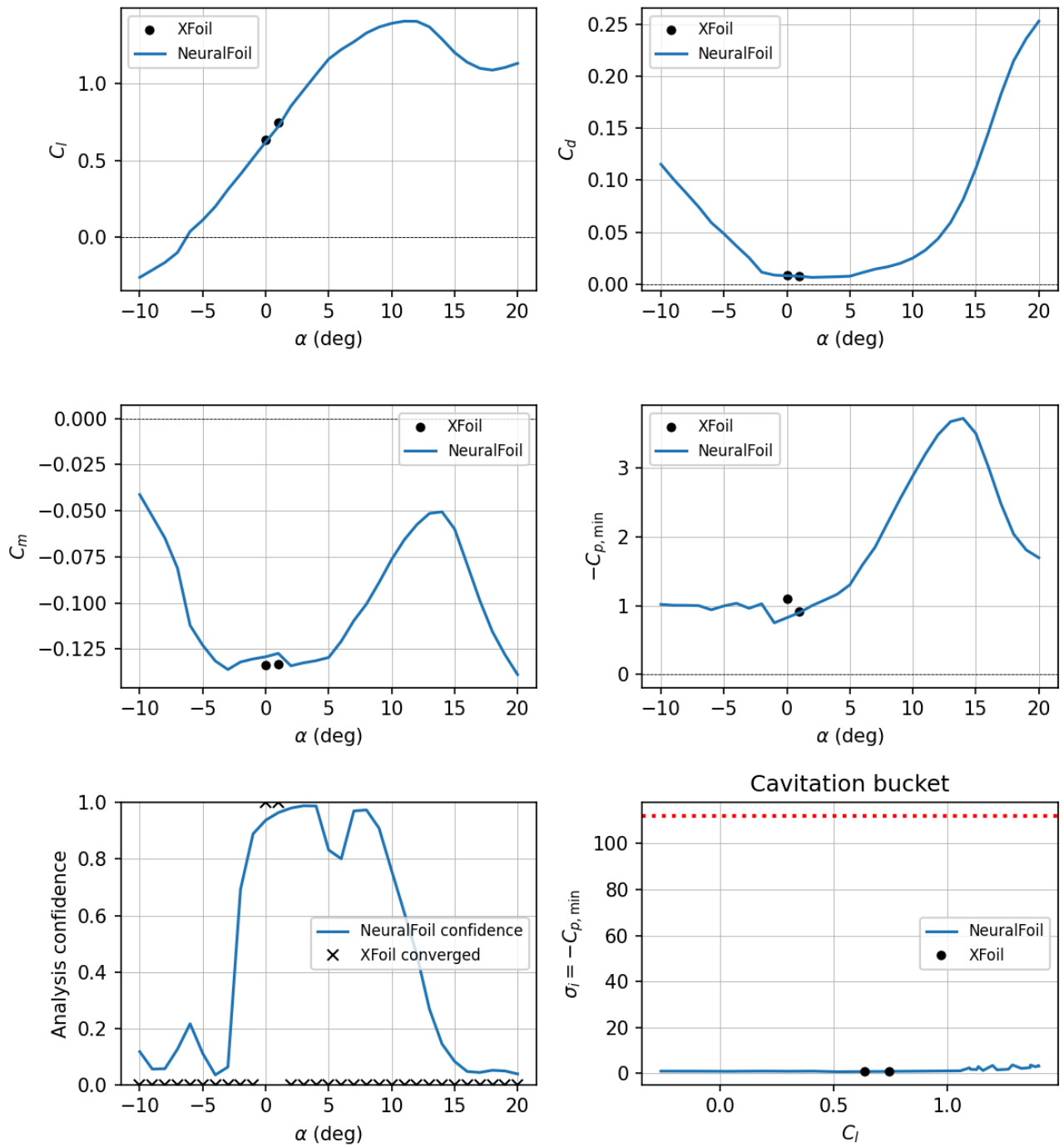
Reconstructed physical outlines of the 20 worst-converging perturbed shapes, ascending.

PGA pairs (535 foils below 0.75 Xfoil convergence)



PGA coordinates of the 535 shapes below the 0.75 threshold — these cluster away from the baseline markers, consistent with the perturbation reaching geometrically implausible extremes at the tails of the sampling distribution.

p7582 — XFoil vs NeuralFoil Re=1e+06 N_crit=7



p7582, the worst-converging shape (34.4% XFoil convergence), at $Re=1e6$, $N_{crit}=7$. XFoil points drop out near stall; the NeuralFoil line and confidence trace are shown alongside for reference.